

SwasthyaSetu — Software Requirements Specification (SRS)

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System Classification: Digital Public Health Infrastructure & Assisted Healthcare Delivery System

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1. Executive Summary

SwasthyaSetu ("Health Bridge") is an assisted digital healthcare delivery platform designed to bridge the structural gap between India's central/state health assurance schemes and the vulnerable citizens entitled to their protection. Rather than functioning as a passive informational portal, SwasthyaSetu operates as an active, human-in-the-loop operational ecosystem linking three core stakeholders:

1. **Citizens:** Vulnerable individuals and household decision-makers seeking coverage, entitlement evaluation, and guided assistance.
2. **ASHA Workers (Accredited Social Health Activists):** Frontline community healthcare facilitators managing household caseloads, assisting with document verification, conducting physical home visits, and navigating enrolment blockers.
3. **Health Administrators:** Supervisory medical and district officers maintaining policy compliance, overseeing ASHA caseload allocations, validating scheme guidelines, and monitoring operational telemetry.

SwasthyaSetu integrates an authoritative, deterministic rule engine that evaluates family-level eligibility across schemes such as **Ayushman Bharat (AB-PMJAY)**, **Janani Suraksha Yojana (JSY)**, **Janani Shishu Suraksha Karyakram (JSSK)**, and state programs like **Ayushman Bharat - Arogya Karnataka (AB-ARK)**.

To overcome severe digital and linguistic literacy barriers in rural and peri-urban demographics, SwasthyaSetu incorporates a **two-tier multilingual delivery architecture**:

- **Trilingual Web Interface:** Full parity across English (`en`), Kannada (`kn`), and Hindi (`hi`).
- **Interactive Multilingual Voice Assistant via Real PSTN:** Powered by **Exotel Telephony** and **Sarvam AI** (`saaras:v3` STT and `bulbul:v3` TTS at 8kHz μ -law), supporting full interactive voice calls over regular non-smartphone mobile networks in Kannada (`kn-IN`), Hindi (`hi-IN`), and English (`en-IN`).

The platform guarantees strict privacy, client-side encryption of authentication secrets, zero leakage of personal health identifiers (PHI) over voice streams, and deterministic governance: AI never hallucinated medical or eligibility rules, acting solely as a speech transcription, speech synthesis, and contextual explanation interface.

2. Problem Statement

Across India, hundreds of millions of low-income citizens qualify for life-saving welfare and healthcare schemes funded by central and state governments. In practice, utilization remains severely constrained due to five systemic friction points:

1. **Complex, Disjointed Scheme Rules:** Eligibility rules depend on intersecting criteria—age thresholds (e.g., universal PM-JAY for senior citizens 70+ irrespective of income), socio-economic categories (BPL, Antyodaya Anna Yojana, NFSA), maternal status, infant age, and state residency. Families struggle to determine who in their household qualifies for what benefit.
2. **The "Last-Mile" Documentation Gap:** A family might qualify in theory, but lacks knowledge of exact mandatory documentation (e.g., Mother & Child Protection [MCP] card, Aadhaar e-KYC on the Ayushman app, or linked bank passbooks). Incomplete paperwork causes hospital turnaways during emergency admissions.
3. **Frontline ASHA Worker Overload:** Frontline ASHA workers are burdened with paper registers, fragmented home visit schedules, missed follow-ups for high-risk maternal patients, and lack of real-time visibility into which households face critical coverage gaps.
4. **Administrative Blind Spots:** District health administrators have no centralized telemetry to assess ASHA caseload distribution, overdue patient interactions, or unresolved field enrollment blockers.
5. **Digital & Language Barriers:** Over 60% of rural beneficiaries lack smartphones or high-speed data access, and cannot read English or formal administrative Hindi/Kannada text. Traditional web-only portals fail completely for illiterate or non-smartphone populations.

SwasthyaSetu addresses these challenges by uniting deterministic eligibility computation, frontline ASHA field case tracking, and standard telephone (PSTN) voice interaction in regional mother tongues.

3. System Objectives

Citizen Objectives

- **Household-Centric Evaluation:** Enable citizens to register their household and family members once, instantly unlocking deterministic eligibility evaluations for every member across all supported schemes.
- **Actionable Guidance:** Provide transparent checklists of required documents, enrollment steps, and official scheme citations rather than generic advice.
- **Direct ASHA Linkage:** Allow citizens to discover, link with, and request direct assistance from their local designated ASHA worker.
- **PSTN Voice Accessibility:** Allow citizens to request a call back or dial in via traditional feature phones to query their household eligibility and case status in Kannada, Hindi, or English without requiring an active internet connection.

ASHA Worker Objectives

- **Digital Caseload Management:** Provide a consolidated, offline-friendly mobile workspace replacing physical registers with digital household records.
- **Attention Signals & Gap Triage:** Automatically flag households with urgent healthcare gaps (e.g., unvaccinated infants, unregistered pregnancies, unassisted 70+ seniors).
- **Assistance Request Workflow:** Enable workers to review, accept, or decline citizen assistance requests, automatically converting accepted requests into tracked field cases.
- **Structured Field Follow-Ups:** Maintain an integrated scheduling system categorizing home visits into `Due`, `Today`, `Overdue`, `Upcoming`, and `Completed`.

Administrator Objectives

- **Real-Time Caseload Oversight:** Monitor total registered households, active cases, overdue follow-ups, and ASHA worker workloads across administrative jurisdictions.
- **Workforce Roster & Assignment:** Inspect ASHA jurisdictions, active caseload counts, and reassign stalled or overloaded cases between workers.
- **Scheme & Rule Governance:** Maintain authoritative scheme metadata, source legal citations, document requirements, and rule definitions with strict versioning.
- **Auditability & Telemetry:** Inspect full system telemetry, including automated n8n event dispatches, PSTN call records, and security audit logs.

Platform & Technical Objectives

- **Deterministic Truth Boundary:** Healthcare eligibility and assistance decisions must be executed by validated rule engines and relational services, never delegated to non-deterministic LLMs.
- **Strict Role-Based Access Control (RBAC):** Cryptographic verification of Firebase Bearer tokens on every API call, enforcing separation between Citizen, ASHA, and Admin roles.
- **Zero Secret Exposure:** Zero exposure of server secrets (`SARVAM_API_KEY`, `EXOTEL_API_TOKEN`, `FIREBASE_PRIVATE_KEY`, `TAVILY_API_KEY`) to clients, telemetry, or public config endpoints.
- **Audio Telephony Invariance:** Deliver G.711 μ -law compliant 8000Hz audio chunked in precise 20ms (160-byte) frames over full-duplex WebSockets for seamless carrier telephony.

4. System Scope

In Scope (Fully Implemented & Verified)

- Single-page responsive web applications for Citizen, ASHA, and Administrator roles built with Next.js 16 (React 19).
- Fastify 5 REST API backend with native WebSocket support (`@fastify/websocket`).
- Deterministic scheme evaluation engine seeded with 5 authoritative schemes: `ab-pmjay`, `jsy`, `state-health-assurance`, `jssk`, and `ab-ark-karnataka`.
- Household and family member lifecycle management (CRUD, health indicators, socio-economic classifications).
- Frontline ASHA caseload workspace: Case detail inspection drawer, task tracking, follow-up scheduling, and field registrations.
- Administrator control center: Caseload oversight, ASHA roster, scheme registry, and system health monitors.

- Full UI localization in English (`en`), Kannada (`kn`), and Hindi (`hi`).
- Full interactive PSTN voice system using Exotel telephony, Sarvam Saaras STT (`saaras:v3`), and Sarvam Bulbul TTS (`bulbul:v3`) in `en-IN` , `kn-IN` , and `hi-IN` .
- Automated voice VAD (Voice Activity Detection), G.711 μ -law transcoding, and 160-byte frame chunking.
- Multi-turn conversational healthcare assistant (Gemini 2.5 Flash) grounded in verified household data.
- Asynchronous domain event automation architecture dispatching to n8n webhook orchestrators.
- Google Cloud Firestore NoSQL storage with strict subcollection hierarchies and security rules.

Out of Scope (Explicitly Excluded / Non-Goals)

- **Direct Clinical Diagnosis or Prescription:** SwasthyaSetu does not diagnose illnesses, interpret medical imagery, or prescribe medications.
- **Direct Financial Disbursement / Banking Integration:** The system tracks eligibility for DBT (Direct Benefit Transfer) but does not execute direct bank transfers or integrate with PFMS/NPCI.
- **Hospital Bed Management:** The system does not manage real-time ICU/hospital bed occupancies.
- **SMS/WhatsApp Gateway Broadcasting (Native):** Handled externally via webhook automations (e.g. n8n); no direct SMPP SMS modem is managed by the core Fastify daemon.
- **Languages Outside en-IN, kn-IN, hi-IN:** Other regional languages (e.g., Telugu, Tamil, Marathi) are intentionally deferred to future extension phases.

5. User Roles & Permission Matrix

The platform enforces three strictly separated roles defined in `shared/types/auth.ts` :

1. `CITIZEN` : A household representative or individual seeking healthcare entitlements.
2. `ASHA` : A frontline health worker authorized to manage a local community caseload.
3. `ADMIN` : A health department supervisor or system administrator with platform-wide oversight.

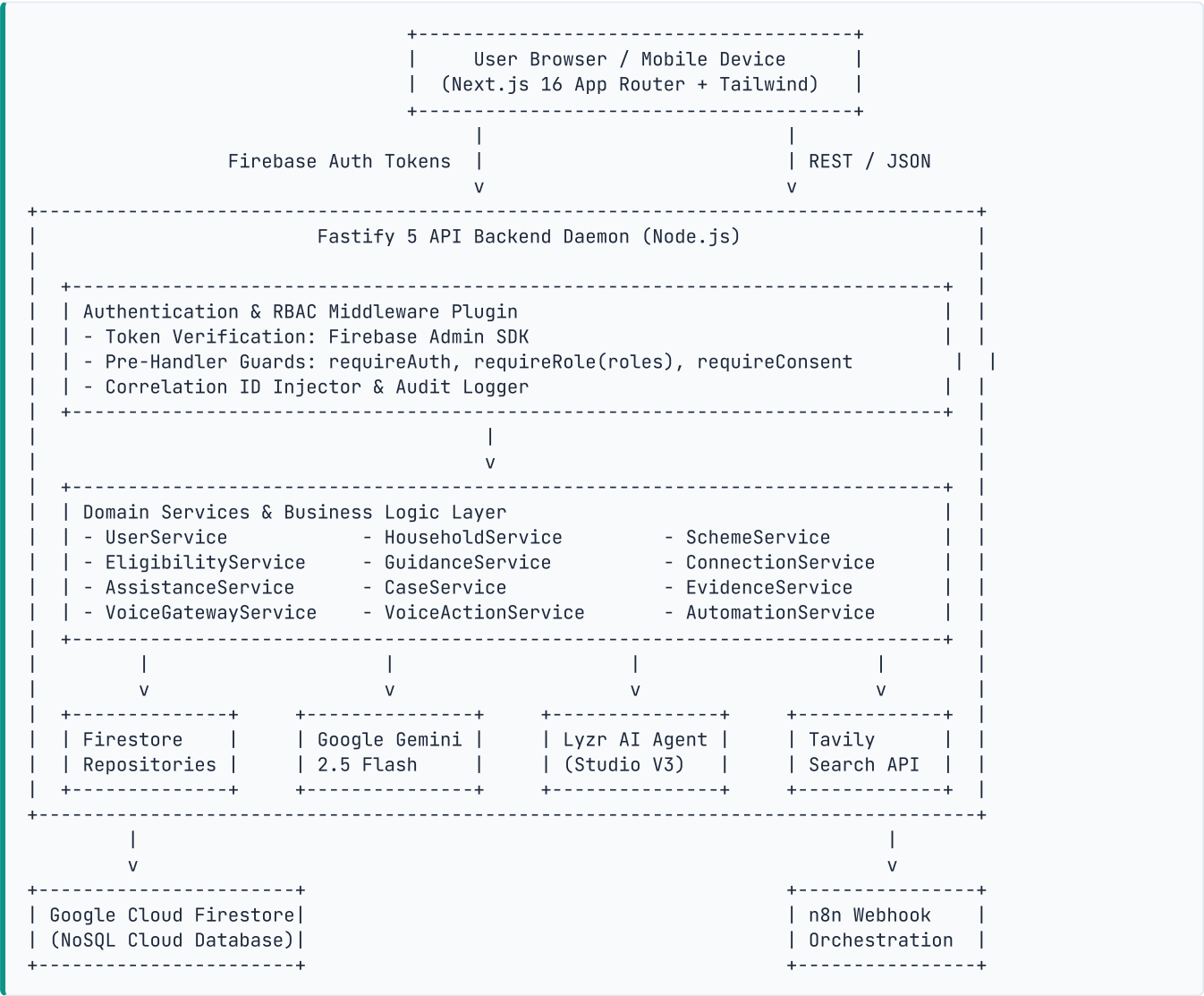
Role Permission Matrix

Functional Capability	Citizen	ASHA Worker	Administrator	Enforcement Mechanism
Browse Public Schemes & Citations	Yes	Yes	Yes	Public Endpoint (<code>/v1/schemes</code>)
Register / Authenticate Account	Yes	Yes (Secret Required)	Yes (Secret Required)	<code>/v1/auth/prevalidate</code> & <code>/v1/auth/register</code>
View Own User Profile (<code>/auth/me</code>)	Yes	Yes	Yes	<code>requireAuth</code> Guard
Submit Data Consent	Yes	Yes	Yes	<code>requireAuth</code> + <code>/v1/auth/consent</code>
Create / Manage Own Household Profile	Yes	No	No	<code>requireRole(["CITIZEN"])</code>
Add / Edit / Remove Own Family Members	Yes	No	No	<code>requireRole(["CITIZEN"])</code>
View Own Household Eligibility & Guidance	Yes	No	No	<code>requireRole(["CITIZEN"])</code>
Search Public ASHA Directory	Yes	Yes	Yes	Authenticated Route
Link to ASHA via Service Code	Yes	No	No	<code>requireRole(["CITIZEN"])</code>
Submit Scheme Assistance Request	Yes	No	No	<code>requireRole(["CITIZEN"])</code>
Request PSTN Callback (<code>/voice/citizen/request-call</code>)	Yes	No	No	<code>requireRole(["CITIZEN"])</code>
Chat with AI Assistant Drawer	Yes	Yes	Yes	<code>requireAuth</code> + Role-Scoped Context
View Assigned Community Caseload	No	Yes	Yes	<code>requireRole(["ASHA", "ADMIN"])</code>
Field-Register New Household	No	Yes	Yes	<code>requireRole(["ASHA", "ADMIN"])</code>
Review / Accept / Decline Assistance Requests	No	Yes	Yes	<code>requireRole(["ASHA", "ADMIN"])</code>
Create / Update Case Tasks & Notes	No	Yes	Yes	<code>requireRole(["ASHA", "ADMIN"])</code>
Schedule / Reschedule / Complete Follow-Ups	No	Yes	Yes	<code>requireRole(["ASHA", "ADMIN"])</code>
Trigger Outbound ASHA Call to Beneficiary	No	Yes	Yes	<code>requireRole(["ASHA", "ADMIN"])</code>
Reassign Cases Between ASHA Workers	No	No	Yes	<code>requireRole(["ADMIN"])</code>
View All Platform Follow-Ups & Telemetry	No	No	Yes	<code>requireRole(["ADMIN"])</code>
Assign Platform Roles to Users	No	No	Yes	<code>requireRole(["ADMIN"])</code>
Query External Evidence via Tavily Search	No	No	Yes	<code>requireRole(["ADMIN"])</code>
Resolve Knowledge Conflicts	No	No	Yes	<code>requireRole(["ADMIN"])</code>

Functional Capability	Citizen	ASHA Worker	Administrator	Enforcement Mechanism
View Voice Telemetry & Automation Health	No	No	Yes	<code>requireRole(["ADMIN"])</code>

6. High-Level System Architecture

6.1 Web Application & Data Plane Architecture



6.2 Real-Time PSTN Voice & Telephony Architecture




```
| model: bulbul:v3 |
| target_language: kn/hi/en-IN |
| sample_rate: 8000Hz |
+-----+
|
| Synthesized 8kHz Audio WAV
v
+-----+
| audio-codec.ts |
| - Extract 16-bit Linear PCM |
| - Transcode via G.711 Table |
| - Chunk into 160-byte (20ms) |
| μ-law (PCMU) Frames |
+-----+
|
| Base64 Encoded Media Frames
v
+-----+
| WebSocket → Exotel → Caller |
+-----+
```

7. Complete Technology Stack

Component Layer	Technology / Package	Verified Version	Purpose in SwasthyaSetu
Frontend Framework	Next.js (App Router)	16.3.3	React 19 server/client hybrid rendering, responsive routing, and asset serving
UI Library	React	19.0.0	Declarative user interface components and state hooks
CSS & Styling	Tailwind CSS (with PostCSS)	4.3.3	Custom utility design system, responsive breakpoints, and custom form primitives
Client Icons	Lucide React	1.34.0	Consistent iconography across dashboards, drawers, status badges, and buttons
Client Validation	Zod	4.4.3	Client-side input validation for registration, forms, and modal interactions
Client Auth SDK	Firebase Web SDK	12.18.0	Client authentication, token lifecycle, and session maintenance
Backend Runtime	Node.js	v22.x	Modern server-side asynchronous execution environment
Backend Framework	Fastify	5.2.1	High-throughput, low-overhead HTTP web framework with robust plugin lifecycle
WebSocket Engine	@fastify/websocket / ws	11.3.0 / 8.21.3	Low-latency duplex communication engine handling Exotel telephony audio frames
CORS Middleware	@fastify/cors	10.1.0	Cross-Origin Resource Sharing security configuration for web clients
Schema Validation	Zod	3.24.2	Runtime API payload validation and schema contracts shared across frontend/backend
Backend Auth SDK	Firebase Admin SDK	13.10.0	Cryptographic verification of Firebase ID tokens and server-side profile syncing
Database	Google Cloud Firestore	Native API	Serverless NoSQL document database storing users, households, cases, and sessions
Conversational AI	@google/genai (Gemini)	2.19.0	Grounded multi-turn contextual assistance via model gemini-2.5-flash
Voice STT	Sarvam AI (Saaras)	saaras:v3	High-accuracy Indian English, Kannada, and Hindi speech-to-text transcription
Voice TTS	Sarvam AI (Bulbul)	bulbul:v3	Natural regional speech synthesis at 8000Hz (Speaker: shubh)
Telephony Gateway	Exotel PSTN & Stream Applet	Cloud REST / WSS	Carrier-grade PSTN inbound/outbound call bridging with real-time WebSocket streaming
Search / Provenance	Tavily Search API	REST v1	Automated discovery and verification of official government scheme circulars
External Agent AI	Lyzyr AI Agent Studio	REST v3	Secondary health entitlement gap analysis and document readiness evaluation
Workflow Automation	n8n Workflow Engine	Webhooks	Asynchronous event dispatching, follow-up polling, and overdue escalations
Testing Framework	Vitest	3.0.6	Comprehensive automated unit, integration, route, and audio codec test suite

Component Layer	Technology / Package	Verified Version	Purpose in SwasthyaSetu
TypeScript	TypeScript Compiler	5.7.3	Strict type safety across frontend, backend, and shared domain models

8. Frontend Architecture

The frontend is organized under the Next.js 16 App Router hierarchy (`/frontend/app/`):

```

frontend/
├── app/
│   ├── layout.tsx           # Root HTML layout, font bindings, I18nProvider, AuthProvider
│   ├── page.tsx            # Public landing page with role-selection & scheme overview
│   ├── globals.css         # Custom CSS properties, theme design tokens, animations
│   ├── error.tsx          # Error boundary component
│   ├── loading.tsx        # Global route transition loader
│   ├── not-found.tsx      # 404 handler
│   ├── unauthorized/page.tsx # Explicit 403 access denial screen
│   ├── auth/
│   │   ├── sign-in/page.tsx # Multi-role sign-in & privileged registration tab
│   │   └── consent/page.tsx # Mandatory data consent acceptance gate
│   ├── citizen/page.tsx    # Citizen Dashboard (Household, Family, Schemes, Actions)
│   ├── asha/page.tsx       # ASHA Workspace (Caseload, Requests, Follow-Ups, Drawer)
│   └── admin/page.tsx      # Admin Console (Oversight, Workforce, Scheme Registry)
├── components/
│   ├── assistant/         # HealthcareAssistantDrawer (Gemini chat)
│   ├── auth/              # ProtectedRoute wrapper with RBAC & consent validation
│   ├── dev/               # DevDiagnosticPanel (environment debug toggles)
│   ├── i18n/              # LanguageSelector dropdown
│   ├── layout/            # AuthenticatedShell, Header, Footer, MobileNav
│   ├── ui/                # Badge, Button, Card, Modal, Input, Textarea, StatusBadge
│   └── voice/              # CitizenCallModal, AshaCallModal (PSTN trigger modals)
├── config/                # env.ts, site.ts
├── hooks/                 # Custom UI & interaction hooks
├── i18n/                  # I18nContext, translation dictionaries (en.ts, kn.ts, hi.ts)
└── services/              # Strongly-typed API client services wrapping fetch

```

Complete Route Inventory

Route Path	Access Boundary	Purpose & Major Sections	Consumed Backend APIs
/	Public	Landing page; hero banner, scheme overview, feature breakdown, role quick-start cards, language toggle	GET /v1/schemes
/auth/sign-in	Public (Unauthenticated)	Role-based sign-in tabs (Citizen, ASHA, Admin); password/phone credentials; registration secret input	POST /v1/auth/prevalidate , POST /v1/auth/sync , POST /v1/auth/register
/auth/consent	Authenticated (Pending Consent)	Statutory healthcare data consent agreement; terms of data sharing and privacy policy	POST /v1/auth/consent , GET /v1/auth/me
/citizen	Role: CITIZEN + Consented	Citizen Dashboard: Household profile card, Family members table, ASHA worker connection status, Recommended scheme cards, Next Steps priority action list	GET /v1/households/me , GET /v1/eligibility/me , GET /v1/guidance/me , GET /v1/connections/citizen/status , POST /v1/assistance/requests , POST /v1/voice/citizen/request-call
/asha	Role: ASHA + Consented	ASHA Workspace: Metric counters, Attention signal triage cards, Assigned household roster, Pending assistance requests, Follow-up scheduling list, Case inspection drawer	GET /v1/cases/my-cases , GET /v1/cases/summary , GET /v1/cases/follow-ups/summary , GET /v1/assistance/requests/asha , PATCH /v1/cases/:id/tasks/:taskId/status , POST /v1/voice/asha/call-citizen
/admin	Role: ADMIN + Consented	Admin Console: Platform metric KPIs, Complete household directory, ASHA workforce distribution table, Platform-wide case oversight, Scheme registry inspector, System health monitors	GET /v1/admin/cases , GET /v1/admin/follow-ups , GET /v1/admin/automation/health , GET /v1/admin/voice/telemetry , POST /v1/admin/cases/assign
/unauthorized	Authenticated	Explicit access denial screen displayed when an authenticated user attempts to access a route outside their RBAC privileges	None

9. Landing Page & Public Experience

The public landing page (`/frontend/app/page.tsx`) serves as the entry point:

- 1. Top Navigation Bar:** Contains the SwasthyaSetu branding, quick-navigation links (*About, Schemes, How It Works*), active Language Selector (English, ಕನ್ನಡ, हिन्दी), and the Primary Authentication CTA (*Sign In / Register*).
- 2. Hero Section:** Clear value proposition highlighting universal healthcare entitlement discovery, assisted delivery through local ASHA workers, and feature-phone voice accessibility.
- 3. Role Quick-Start Cards:**
 - *Citizen Card:* Highlights instant household eligibility checks and direct ASHA support.
 - *ASHA Worker Card:* Directs frontline health workers to caseload management and task tracking.
 - *Administrator Card:* Directs supervisory officers to the system oversight console.
- 4. Verified Scheme Showcase:** Interactive card grid rendering authoritative schemes fetched dynamically from `GET /v1/schemes`, displaying benefit caps (e.g., ₹5,00,000 for PM-JAY), target demographics, and required documentation.
- 5. Interactive Helpline Teaser:** Displays the national toll-free helpline number and explains the feature-phone callback mechanism.
- 6. Active Session Detection:** If a user is already signed in with a valid token, the hero CTA dynamically converts from *Sign In* to *Go to Your Dashboard*, routing them to `/citizen`, `/asha`, or `/admin` based on their resolved role.

10. Authentication, Authorization & Consent Flow

10.1 Multi-Stage Authentication Flow

1. Citizen/Staff Enters Credentials on `/auth/sign-in`
|
v
2. If Privileged (ASHA or Admin):
Client calls `POST /api/v1/auth/prevalidate` with `{ requestedRole, registrationSecret }`
Backend compares secret against SHA-256 hash environment variable
|
+→ If Invalid Secret: Rejects immediately with 401 Unauthorized
|
v
3. Firebase Client SDK Authenticates User (Email/Password or Phone Auth)
Returns Firebase ID Token (JWT)
|
v
4. Client sends Bearer Token to `POST /api/v1/auth/sync` or `/register`
|
v
5. Fastify Auth Plugin verifies token with Firebase Admin SDK
 - Inspects `decodedToken.uid`
 - Retrieves or creates Firestore record in `/users/{uid}`
 - Assigns validated role (CITIZEN default, or ASHA/ADMIN if verified)|
v
6. User Profile returned to Client with `isConsentRequired` flag
|
+→ If `isConsentRequired = true`: Redirect to `/auth/consent`
+→ If `isConsentRequired = false`: Redirect to Role Portal (`/citizen`, `/asha`, `/admin`)

10.2 Server-Side Security Guards

Every protected API route in the Fastify backend is guarded by preHandler hooks defined in `backend/src/plugins/guards.ts`:

- `requireAuth`: Verifies the `Authorization: Bearer <token>` header against Firebase Admin. Binds `request.user` (decoded token) and `request.userProfile` (Firestore user record).
- `requireRole(allowedRoles: UserRole[])`: Inspects `request.userProfile.role`. If the role is not included in `allowedRoles`, halts execution immediately with HTTP 403 `INSUFFICIENT_ROLE`.
- `requireConsent`: Checks whether `request.userProfile.consent` is recorded for `CURRENT_CONSENT_VERSION` ("2026.1"). If not accepted, halts with HTTP 403 `CONSENT_REQUIRED`.

11. Citizen Portal Functional Specification

Accessible exclusively to users with role `CITIZEN` at `/citizen`.

11.1 Household Profile Management

- **Household Head Information:** Head of Household full name, contact phone number, and gender.
- **Socio-Economic Classification:** Ration card category selection (`BPL` - Below Poverty Line, `AAY` - Antyodaya Anna Yojana, `APL` - Above Poverty Line, `NON_RATION`).
- **Location Hierarchy:** State, district, taluk/block, village/ward, and PIN code.
- **Ration Card Tracking:** Registered ration card number used for cross-referencing public beneficiary registries and verifying voice caller identity.

11.2 Family Member Roster

- **Member Attributes:** Full name, relationship to head (Self, Spouse, Son, Daughter, Father, Mother, Grandfather, Grandmother), age in years, and gender.
- **Vulnerability & Healthcare Indicators:**
 - `isPregnant`: Boolean flag indicating active pregnancy.
 - `isLactating`: Boolean flag indicating nursing mother (postpartum < 6 months).
 - `hasDisability`: Boolean flag indicating physical or sensory disability.
 - `hasChronicIllness`: Boolean flag for non-communicable disease tracking.
- **Automatic Triggers:** Updating family member attributes immediately invalidates cached eligibility and triggers a re-evaluation across all active schemes.

11.3 ASHA Worker Connection Card

- **Connection Status:** Displays current linkage state (`NONE`, `PENDING`, `ACTIVE`, `REJECTED`).
- **Linked Worker Details:** If connected, displays the designated ASHA worker's full name, assigned service code (e.g., `ASHA-BLR-042`), operational jurisdiction, and official contact phone.
- **Connection Actions:** Allows the citizen to search for an ASHA worker by service code and transmit a connection request, or revoke an existing connection.

11.4 Recommended Schemes & Entitlement Status

- **Dynamic Scheme Cards:** Displays all schemes for which at least one household member is determined `ELIGIBLE` or `NEEDS_MORE_INFO`.
- **Card Content:** Scheme official name, authority, maximum financial coverage (e.g. ₹5 Lakhs for PM-JAY, ₹1,400–₹6,000 for JSY), qualifying household members, required document checklist, and official policy source citation.
- **Request ASHA Assistance CTA:** Directly embedded button on every eligible scheme card allowing the citizen to request physical home assistance from their linked ASHA worker with one click.

11.5 Next Steps & Priority Action Checklist

- **Deterministic Action Prioritization:** Consolidates all required enrollment steps across eligible schemes into an ordered action list (e.g., *"Complete 70+ Senior Citizen e-KYC on Ayushman App for Eerappa"*).
- **Status Badges:** Clearly delineates items requiring citizen action versus items pending ASHA verification.

11.6 PSTN Callback Modal & Voice Assistant

- **PSTN Callback Trigger:** Embedded modal (`CitizenCallModal`) allowing the citizen to confirm their phone number and preferred language (English, Kannada, Hindi) to request an immediate outbound phone call from the SwasthyaSetu voice assistant.
- **In-Browser Gemini Assistant Drawer:** Slide-out drawer (`HealthcareAssistantDrawer`) allowing the citizen to type or speak questions in natural language, receiving responses strictly grounded in their household's actual members and eligibility.

12. ASHA Workspace Functional Specification

Accessible to users with role `ASHA` or `ADMIN` at `/asha`.

12.1 Operational Metric Dashboard

- **Total Assigned Caseload:** Total count of registered households under the worker's jurisdiction.
- **Active Cases:** Number of households with ongoing assistance workflows.
- **Attention Signals:** Number of households with unaddressed health entitlement gaps.
- **Due Follow-Ups:** Number of scheduled field visits due today or overdue.

12.2 Attention Signal Triage

- **Proactive Gap Detection:** Evaluates household profiles to highlight vulnerable conditions:
 - Senior citizens (70+) lacking Ayushman Vay Vandana card generation.
 - Pregnant mothers without registered MCP cards or JSY institutional delivery linkages.
 - Low-income households with pending document submissions.
- **One-Click Case Initiation:** Allows the ASHA worker to convert an attention signal directly into an active tracked case.

12.3 Caseload Directory & Search

- **Search & Filters:** Real-time filtering of assigned households by family head name, ration card number, or village.
- **Household Health Cards:** Compact cards rendering family size, head of household, income category, active cases, and contact details.
- **Call Citizen Action:** Embedded modal (`AshaCallModal`) allowing the worker to trigger an assisted telephony call to the beneficiary household.

12.4 Assistance Request Management

- **Citizen Request Inbox:** Lists incoming assistance requests submitted by citizens via their portal or telephony calls.
- **Workflow Actions:**
 - *Accept Request:* Binds the request to the household's active case (or creates a new case) and initializes standard milestone tasks.
 - *Decline Request:* Records a mandatory administrative explanation note and notifies the citizen.
 - *Resolve Request:* Marks the workflow completed upon successful benefit receipt.

12.5 Case Detail Inspection Drawer

Clicking any case opens the comprehensive Case Detail Drawer containing five functional tabs:

1. **Journey & Milestones:** Visual progress tracker showing the current journey stage (e.g., `DOCUMENT_COLLECTION` → `VERIFICATION` → `ENROLMENT` → `CARD_ISSUANCE` → `BENEFIT_DELIVERY`).
2. **Tasks Checklist:** Granular actionable tasks (e.g., "Collect Aadhaar Card of Grandfather", "Submit MCP card copy to PHC"). Workers can toggle tasks between `PENDING` , `IN_PROGRESS` , `COMPLETED` , and `BLOCKED` .
3. **Follow-Up Visits:** Interface to schedule upcoming home visits with a specific date, time, and visit agenda.
4. **Field Notes:** Chronological audit log of notes recorded by the worker during home visits.
5. **Beneficiary Profile:** Household roster and deterministic scheme eligibility summary.

12.6 Field Registration Flow

- **In-Field Household Onboarding:** Allows ASHA workers to register new households encountered during field surveys directly from their workspace, capturing ration details, family members, and health indicators without requiring prior citizen self-registration.

13. Admin Console Functional Specification

Accessible exclusively to users with role `ADMIN` at `/admin` .

13.1 Platform Overview Dashboard

- Aggregates platform-wide Key Performance Indicators (KPIs): Total Registered Households, Total ASHA Workforce, Active Caseloads, Total Completed Follow-Ups, and System Operational Health.

13.2 Complete Household Directory

- Full-text search and multi-parameter filtering across all registered households in the database.

- Provides supervisory inspection of household members, assigned ASHA workers, and active cases.

13.3 ASHA Workforce Management

- Tabular roster of all registered ASHA workers, displaying their unique service codes, assigned jurisdictions (state, district, taluk), active case counts, and overdue visit ratios.
- **Case Reassignment:** Administrative modal to reassign households or individual cases from an overloaded or absent worker to another active worker.

13.4 Platform-Wide Case Oversight

- Real-time audit stream of all cases across the system.
- Ability to filter by priority (`URGENT` , `HIGH` , `NORMAL` , `LOW`) and status (`BLOCKED` , `NEEDS_ATTENTION` , `IN_PROGRESS`).
- Direct access to inspect journey blockers, overdue follow-ups, and field notes.

13.5 Scheme Registry & Policy Governance

- Comprehensive inspector for all registered health schemes and ruleset versions.
- Displays administrative authority, benefit summaries, required document lists, and official government policy citations.

13.6 System Monitoring & Telemetry

- **API & Service Health:** Live connection status for Firestore, Sarvam AI, Exotel Telephony, and n8n Webhooks.
- **Voice Telemetry:** Call completion rates, turn distribution metrics, average latency per speech turn, and error rate tracking.
- **Automation Orchestration Telemetry:** Live health status of the n8n webhook dispatcher, displaying recent domain event dispatches, payload deliveries, and retry states.

14. Healthcare Scheme & Deterministic Eligibility Engine

SwasthyaSetu implements an authoritative, deterministic rule engine (`backend/src/services/eligibility/rule-engine.ts`) to ensure zero hallucinations in welfare allocation.

14.1 Verified Production Schemes

1. Ayushman Bharat - PMJAY (ab-pmjay)
Authority: National Health Authority (NHA), MoHFW, Government of India
Benefit: Up to ₹5,00,000 yearly hospitalization coverage per family across empaneled hospitals.
Pathways:
a) Universal 70+ Senior Citizen Pathway: Any citizen aged ≥ 70 years is deterministically ELIGIBLE regardless of income, caste, or socio-economic criteria.
b) Socio-Economic Pathway: Households categorized as BPL or AAY are ELIGIBLE.
Required Documents: Aadhaar Card (with e-KYC on Ayushman App), Ration Card.
2. Janani Suraksha Yojana - JSY (jsy)
Authority: Ministry of Health and Family Welfare (MoHFW), Government of India
Benefit: Cash assistance (₹1,400 in rural / ₹1,000 in urban) for institutional delivery.
Criteria: Any female household member who is actively pregnant (isPregnant = true) or lactating.
Required Documents: Mother & Child Protection (MCP) Card, Bank Passbook linked to Aadhaar.
3. Janani Shishu Suraksha Karyakram - JSSK (jssk)
Authority: Ministry of Health and Family Welfare (MoHFW), Government of India
Benefit: 100% completely free and cashless delivery and pediatric care up to 1 year of age.
Criteria: Active pregnancy, nursing mothers, or infants aged ≤ 1 year.
Required Documents: Hospital Admission Slip, Government Hospital MCP Card.
4. State Health Assurance Scheme (state-health-assurance)
Authority: State Department of Health and Family Welfare
Benefit: Secondary and tertiary hospitalization cover supplementing central schemes.
Criteria: Household possesses valid state residency and BPL / NFSA ration card.
Required Documents: State Ration Card, Income Certificate.
5. Ayushman Bharat - Arogya Karnataka - AB-ARK (ab-ark-karnataka)
Authority: Suvarna Arogya Suraksha Trust (SAST), Government of Karnataka
Benefit: Cashless co-branded health protection for Karnataka residents up to ₹5,00,000.
Criteria: State residency in Karnataka AND valid NFSA Ration Card (BPL/AAY).
Required Documents: Karnataka Ration Card (BPL/AAY), Aadhaar Card.

14.2 Deterministic Evaluation Logic

The evaluation engine processes a household document through the following sequence:

- Household Evaluation:** Evaluates household-level attributes (state of residence, income/ration category).
- Member-Level Evaluation:** Iterates through every family member in the household:
 - Evaluates age boundaries (e.g. `age  $\geq 70$`).
 - Evaluates maternal indicators (`isPregnant`, `isLactating`).
 - Evaluates vulnerability tags (`hasDisability`, `hasChronicIllness`).
- Condition Classification:** Evaluates rules using strict Boolean logic:
 - `ELIGIBLE`: All mandatory criteria satisfied. Returns qualifying member ID, benefit details, and action plan.
 - `NEEDS_MORE_INFO`: Potential entitlement identified, but critical documentation or age/income verification is pending.
 - `NOT_ELIGIBLE`: Deterministic criteria failed. Generates an explicit explanation (e.g., *"Household income category APL does not meet BPL requirements, and no senior citizens aged 70+ reside in the household"*).
- Action Plan Synthesis:** Maps every eligible scheme directly to required documents and concrete next steps.

15. Case Management & Beneficiary Journey Engine

The case engine (`backend/src/services/case.service.ts`) governs the operational lifecycle of assisted delivery:

15.1 Case Lifecycle State Machine



15.2 Case Data Structure & Subcollections

Each case document in `/cases/{caseId}` maintains:

- `householdId` : Reference to target household.
- `assignedAshaUid` : UID of the assigned frontline worker.
- `status` : Current `CaseStatus` enum.
- `priority` : `LOW` , `NORMAL` , `HIGH` , `URGENT` .
- `schemeId` & `beneficiaryMemberId` : Optional linkage to specific scheme and member.
- **Subcollections:**
 - `/cases/{caseId}/tasks` : Ordered checklist of action items.
 - `/cases/{caseId}/followups` : Scheduled home visits and calls.
 - `/cases/{caseId}/notes` : Field visit notes with timestamps and author metadata.
 - `/cases/{caseId}/activities` : Immutable audit trail of every status change and assignment.

16. Assistance Request & Connection Lifecycle

16.1 Citizen-to-ASHA Connection Lifecycle

1. **Search:** Citizen queries public ASHA directory (`GET /v1/connections/directory/ashas`).
2. **Request Link:** Citizen submits connection request using ASHA service code (`POST /v1/connections/citizen/link`). Connection status set to `PENDING` .
3. **ASHA Review:** ASHA reviews pending link requests in their workspace (`GET /v1/connections/asha/requests`).
4. **Accept / Reject:** ASHA accepts (`POST /v1/connections/asha/requests/:id/respond`). Connection becomes `ACTIVE` . The citizen's household is now linked to the ASHA worker's caseload.

16.2 Assistance Request Flow

1. **Citizen Request:** Citizen clicks "Request ASHA Assistance" on a scheme card (`POST /v1/assistance/requests`).
2. **Inbox Notification:** ASHA receives the request in their workspace inbox (`GET /v1/assistance/requests/asha`).
3. **Action:**
 - **Accept** (`POST /v1/assistance/requests/:id/accept`): The system automatically creates a new tracked `AshaCase` (or binds to an existing active case), initializes scheme journey milestones, and links the request.
 - **Decline** (`POST /v1/assistance/requests/:id/decline`): Records the decline reason and notifies the citizen.

17. Follow-Up & Scheduling Subsystem

Follow-ups represent scheduled real-world interactions between frontline workers and beneficiaries.

17.1 Categorization Rules

The system evaluates `dueAt` timestamps relative to the local operational date (`todayStr` in `Asia/Kolkata`):

- **Due Today** : Status is `PENDING` AND date matches current calendar day (`dueDateStr = todayStr`).
- **Overdue** : Status is `PENDING` AND `dueDate < now` AND date is strictly prior to today (`dueDate < now && !isToday`).
- **Upcoming** : Status is `PENDING` AND `dueDate > today` .
- **Completed** : Status is `COMPLETED` (`completedAt` populated).
- **Cancelled** : Status is `CANCELLED` (`cancelReason` populated).

17.2 Actions

- **Mark Complete** (`PATCH /v1/cases/:id/follow-ups/:followUpId/status`): Sets status to `COMPLETED` , records outcome notes, and updates case `lastContactAt` .
- **Reschedule** (